

BTC Options Cycle Strategy

Systematic Deribit Premium Selling — Backtest Report

Confidential | Generated July 2026

Backtest Period: Mar 2021 — May 2026 (5.26 years, 324 trades)

+32.5%	5.90	-8.0%	94.8%
CAGR (BTC)	Sharpe Ratio	Max Drawdown	Win Rate

5.00 BTC → 21.91 BTC | +338% cumulative return (BTC) | +433% in USD incl. BTC appreciation

1. Executive Summary

This report presents the backtest results of a systematic BTC options premium-selling strategy on Deribit. The strategy uses the Bitcoin halving cycle as a directional clock: sell puts in the post-halving bull phase (M0-18), sell calls through the top and into the bear (M18-30), and sell puts again in the late bear (M30-48). All returns are denominated in BTC — pure options trading alpha, no spot beta included.

The strategy was extracted from 69 monthly Deribit statements (Dec 2020 - Jun 2026) and independently validated. The equity-scaled variant compounds returns by sizing positions proportional to account equity (size = equity / 5.0 BTC).

2. Key Performance Metrics

Metric	Fixed Size	Equity-Scaled	Equity + DVOL
Starting Equity	5.00 BTC	5.00 BTC	5.00 BTC
Final Equity	12.80 BTC	21.91 BTC	16.11 BTC
Total P&L	+7.80 BTC	+16.91 BTC	+11.11 BTC
CAGR	+19.6%	+32.5%	+25.0%
Max Drawdown	-7.9%	-8.0%	-4.9%
Sharpe Ratio	4.79	5.90	5.76
Win Rate	94.8%	94.8%	94.8%
Total Trades	324	324	324
Avg IM Usage	14.4%	24.0%	20.6%
Peak IM	31.8%	32.2%	32.7%

3. The Direction Dial — Core Edge

Direction is determined by a single variable: months since the last Bitcoin halving. Both historical tops occurred at M18-19 (Nov 2021, Nov 2025). The clock has been validated across 1.5 complete cycles.

Cycle Month	Phase	Sell Side
M0 - M18	Halving to Top	SELL PUTS
M18 - M30	Top to Bear Belly	SELL CALLS
M30 - M48	Late Bear to Next Halving	SELL PUTS

4. Operations Manual

Strikes (Far Leg): Nearest strike to 2.0x spot (calls) / 0.6x spot (puts). User's own multiples beat theoretical 5-delta targeting.

Strikes (Near Leg): Strike closest to 20-delta. Distance breathes with DVOL: ~+7%/-4% at DVOL 40, ~+13%/-9% at DVOL 80.

Tenor: Far leg: ~6-month tenor, entered monthly. The ladder self-builds (6 anchors rolling). Near leg: weekly (Thursday).

Cadence: Near leg every Thursday. Far anchor first Thursday of the month. Expiries recycle margin on the same rhythm.

Dynamic Sizing: $\text{size_btc} = \text{equity} / 5.0$. With 2.5 BTC equity: trade 0.5. With 5 BTC: trade 1.0. With 10: trade 2.0.

Crash Boost: 2x sizing for 28 days after a 12%+ drawdown from 60-day high. Captures vol spikes when premium is richest. Duration cut from 42d to 28d in Jul 2026 — see section 10.

5. Pause Rules & Guards

Far Leg Pause: Skip when DVOL < 40. In backtest, low-DVOL anchors sold 7 positions for 0.015 BTC total but lost -0.043 BTC (14% breach rate).

Near Leg: Never pause. Profitable in every DVOL bucket including < 40 (+0.059 BTC), because the 20-delta strike slides closer and keeps paying.

Margin Rail: IM ≤ 70% of equity. Above 70%: skip near adds. Never add just because margin is idle. Target steady-state: 20-30% IM.

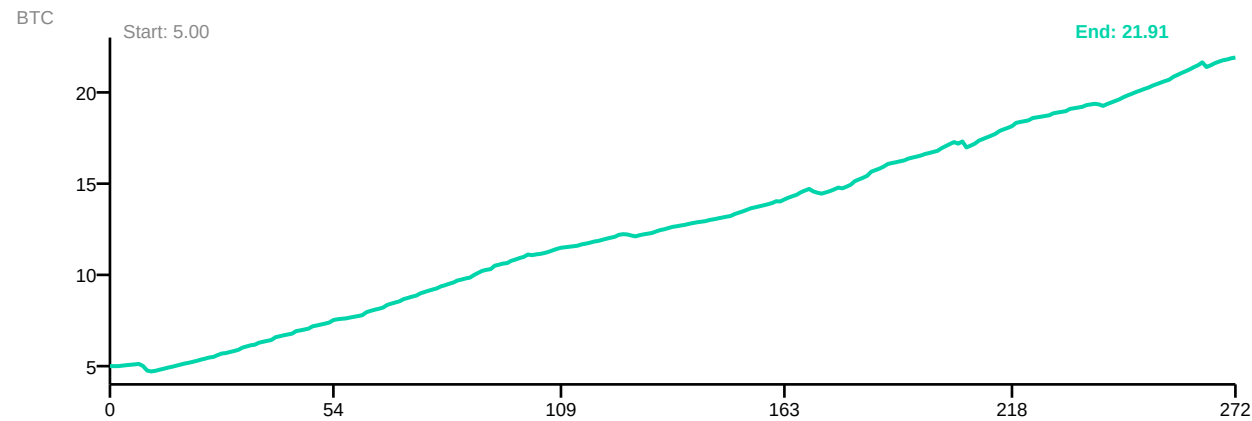
Ignition Guard: Never open a far CALL whose expiry lands in M34-50. This is when long calls return +370-514% — short calls get run over.

6. Yearly Breakdown (Equity-Scaled)

Year	P&L (BTC)	Return %	Trades	Win Rate	Start Eq	End Eq
2021	+2.08	+41.5%	51	94.1%	5.00	7.08
2022	+3.76	+53.1%	62	100.0%	7.08	10.84
2023	+2.47	+22.8%	64	93.8%	10.84	13.31
2024	+3.76	+28.2%	64	92.2%	13.31	17.06
2025	+3.32	+19.5%	62	93.5%	17.06	20.38
2026	+1.53	+7.5%	21	95.2%	20.38	21.91

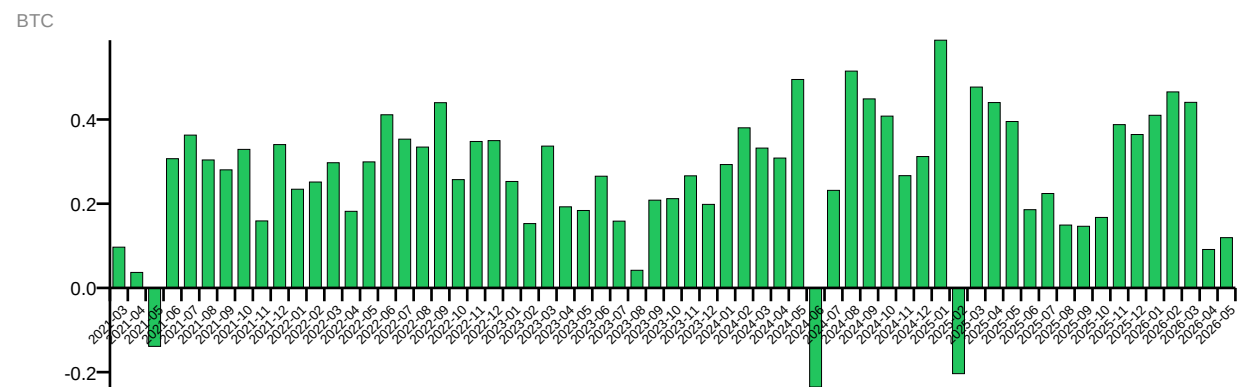
7. Equity Curve (Equity-Scaled, Weekly)

5.00 BTC starting equity, weekly snapshots. Size scales with equity; crash boost 2x after 12%+ drawdowns.



8. Monthly P&L; Distribution

Monthly P&L; in BTC (equity-scaled). Green = profit, Red = loss.



Positive months: 60 | Negative months: 3 | Best: +0.588 BTC | Worst: -0.235 BTC | Average: +0.268 BTC

9. Level 2 Validation (Real Trade Prices)

A second-level backtest replaced Black-Scholes pricing for far legs with REAL Deribit trade prices (quarterly instruments, daily resolution, 14 expiries from 2022-2025). Results:

Metric	Level 1 (BS-only)	Level 2 (Hybrid)
CAGR	+19.5%	+20.7%
Sharpe	3.03	3.46
Max DD	-13.8%	-9.3%
Total BTC	+3.17	+3.46
Far-leg breach	7.0%	0.0%

BS overestimates breaches because it prices with flat DVOL (no skew). Real far-OTM tails are safer.

10. Risk Tuning — Crash Boost Duration (Jul 2026)

The crash boost doubles position size for a fixed window after a 12%+ drawdown. It is a leveraged bet on a fast V-recovery: it pays in a bull regime and costs when the drawdown CONTINUES. The two worst months in the sample (Jun 2024, Feb 2025 — both short-put phases) are exactly that failure mode. The window was cut from 42 to 28 days:

Duration	CAGR	Sharpe	2024 within-yr DD	2025 within-yr DD
21 days	+31.8%	6.04	-2.4%	-2.8%
28 days (live)	+32.5%	5.90	-3.5%	-3.6%
35 days	+33.2%	5.70	-5.9%	-2.8%
42 days (former)	+33.2%	4.97	-6.7%	-8.9%

Never worse on drawdown in any year; twice dramatically better. Cost: -0.7% CAGR. Evidence: a broad plateau (every duration from 10 to 35 days beats 42 — the cliff is at 42, and 21 is not special), a contiguous good region covering 21 of 49 cells on the threshold x duration surface, and 14 distinct crash episodes across 2021/2023/2024/2025/2026. HONEST LIMIT: this change FAILED a pre-registered split-half out-of-sample test and improves only 2 of 6 years on Sharpe. Both gates are structurally biased against risk-reduction changes — a change that trims the fat middle to lift the left tail will always fail them — but this is not validated alpha. It was adopted because -0.7% CAGR is inside the noise band of a 1.5-cycle backtest, making the insurance close to free, NOT because the data compels it. Note the feature overall is a major return driver: crash boost OFF yields 19.6% CAGR vs 32.5% ON. Duration only tunes risk within it.

11. Caveats

- The halving clock is fitted on 1.5 cycles / 2 tops — strong in-sample evidence, not a universal law.
- Projected Apr 2028 halving and 4-year repetition are assumptions, not certainties.
- Month-end snapshots miss intra-month rolling; the simulation holds positions to expiry.
- Level 2 coverage: 42% of far legs have real prices. The 46% BS fallback occurs for extreme OTM strikes.
- v1 of this strategy claimed +25.6% for 5-delta far targeting — that was a solver bound bug (corrected).

12. Current State (Jul 2026)

M26.6 of the Apr 2024 cycle — currently SELLING CALLS. Flip to PUTS at M30 (approx. 20 Oct 2026). From Feb 2027 (M34): no new far calls (ignition guard). From Apr 2027 (M36): long-call window opens. Crash boost is ACTIVE (BTC -15.3% from 60-day high), sizing at 2x until Sep 3, 2026.